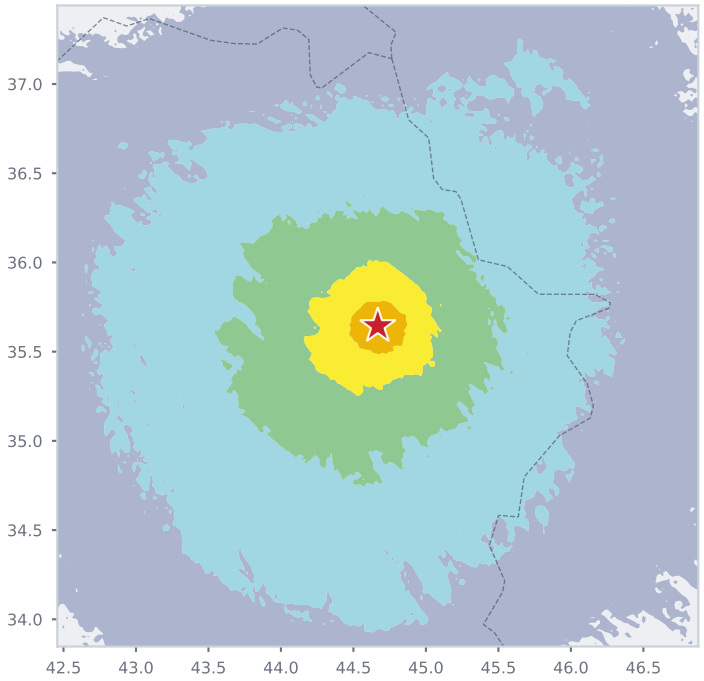
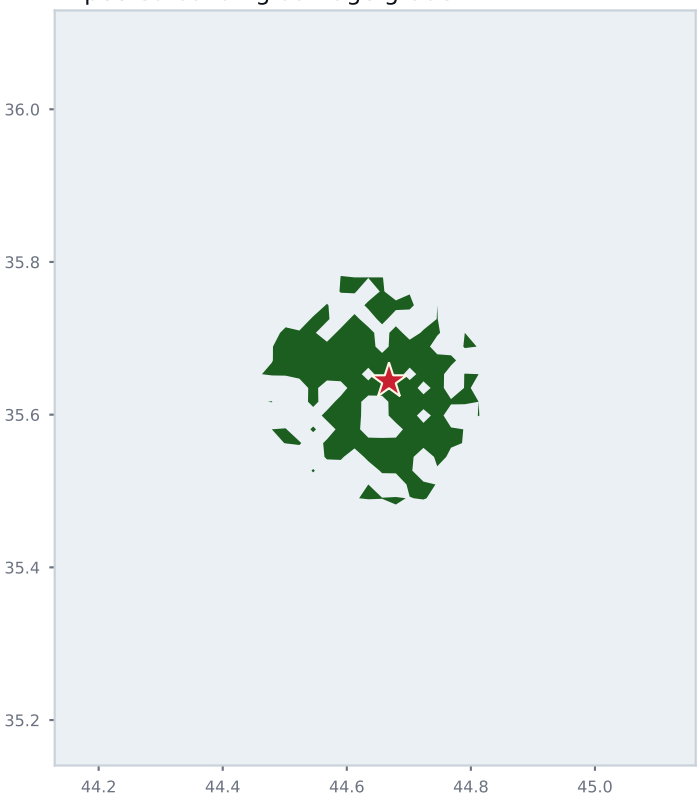
Shaking intensity (IMS-25)




Expected building damage grade




IMS-25 intensity




Expected damage grade (IMS-25)




Damage alert

**ORANGE** Widespread damage; many buildings unusable **Provisional** **Transit occupancy**

Buildings heavily damaged (DG3 or worse)

## 1.4 thousand to 23 thousand

Central estimate 6.1 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

## 3.8 million

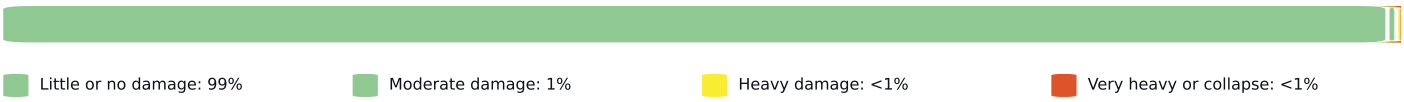
21% of them inferred from built-up area, not observed footprints.

People by shaking level

| | |
|---|---|
| **VIII** | 24 |
| **VII** | 11 thousand |
| **VI** | 1.6 million |
| **V** | 3.7 million |
| **IV** | 6.4 million |

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



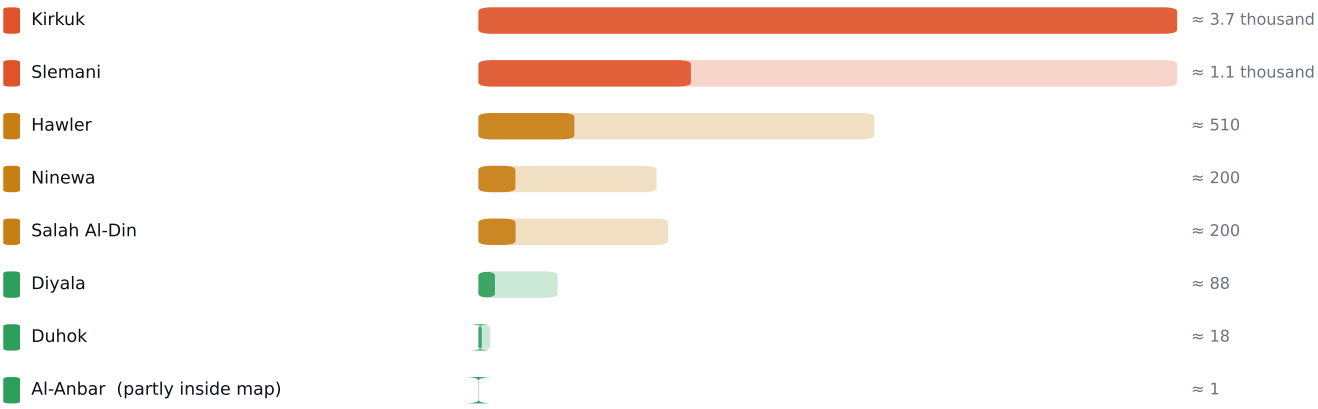
Little or no damage: 99% | Moderate damage: 1% | Heavy damage: <1% | Very heavy or collapse: <1%

## Which buildings failed

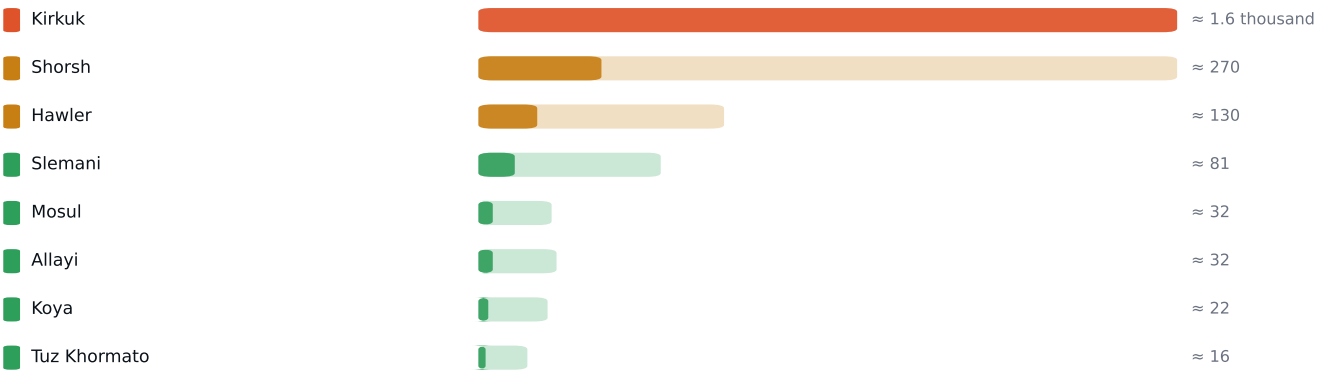
Share of each type's own buildings heavily damaged, worst first

| | | | |
|---|---|---|---|
| Rubble stone, fieldstone | class A | **0.6%** | ≈ 680 buildings |
| Adobe, earth brick | class A | **0.4%** | ≈ 500 buildings |
| Manufactured stone units, timber floors | class B | **0.2%** | ≈ 940 buildings |
| Simple stone | class B | **0.2%** | ≈ 650 buildings |
| RC flat slab and others | class B | **0.1%** | ≈ 4 buildings |
| RC frame — no / low code (ERD-L) | class C | **0.1%** | ≈ 110 buildings |
| Manufactured stone with RC floors | class C | **0.0%** | ≈ 620 buildings |

## Governorates most affected: heavily damaged buildings, central estimate and 90% range




## Cities most affected



## Every governorate inside the map

| Area | Buildings | Heavy (P50) | 90% range | % of its stock |
|---|---|---|---|---|
| Kirkuk | 568 thousand | 2.3 thousand | 830 to 16 thousand | 0.40% |
| Slemani | 669 thousand | 670 | 240 to 4.8 thousand | 0.10% |
| Hawler | 662 thousand | 320 | 130 to 2.1 thousand | 0.05% |
| Ninewa | 792 thousand | 140 | 57 to 950 | 0.02% |
| Salah Al-Din | 561 thousand | 130 | 44 to 1.0 thousand | 0.02% |
| Diyala | 316 thousand | 59 | 23 to 420 | 0.02% |
| Duhok | 259 thousand | 14 | 6 to 63 | 0.01% |
| Al-Anbar | 13 thousand | 1 | none to 4 | 0.01% |

## How this report was made

**Hazard engine** Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

**Observations** 0 seismic stations, 0 felt-report cells (felt channel: none)

**Damage scale** IMS-25

### Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

### What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml19580001, version 2, generated 19 Oct 2026, 11:37 UTC.